

MANIPAL UNIVERSITY JAIPUR
Department of Electronics C Communication Engineering
Course Hand-out

A. Basic Details:

Programme Name:	B. Tech (ECE)
Course Name:	Data structure and Algorithms
Course Code:	ECE2104
LTPC (<i>Lecture Tutorial Practical Credits</i>):	3 0 0 3
Session:	July-Nov 2026
Class:	Second Year
Course Coordinator:	Dr. Kamal Kishor Upadhyay
Course Instructor(s):	❖ Dr. Kamal Kishor Upadhyay ❖ Dr. D.V. Prashant
Additional Practitioner(s) – if any (<i>Industry Fellow/ Visiting Faculty/ Adjunct Faculty, etc.</i>):	Mr. Vipul Vasistya (Senior Associate) Morgan and Stanley

B. Introduction: Data Structures and Algorithms (DSA) provide a systematic study of abstract data types (ADTs), data organization techniques, and algorithmic design principles that enable efficient computation. The course covers the implementation and analysis of linear and non-linear data structures, including linked lists, stacks, queues, trees, heaps, graphs, and hash tables, using C++ or Python. It emphasizes algorithm development for searching, sorting, graph traversal, shortest-path computation, and data manipulation while evaluating their correctness, time complexity, and space complexity. Students develop the ability to select appropriate data structures and algorithms based on computational requirements, memory constraints, and application-specific performance, forming a strong foundation for advanced topics such as operating systems, databases, artificial intelligence, computer networks, and software engineering.

C. Course Outcomes

<i>CO Statement</i>	<i>CO</i>	<i>Level</i>	<i>Target Attainment %</i>	<i>Target Attainment level</i>
Explain concepts of object-oriented programming using C++/Python	1	L2	85	3

Explain the basic operations on arrays, lists, stacks and queue data structures with their programming examples.	2	L3	75	2
Develop the notions of trees, binary search trees, Red Black tree, Heap with their programming examples.	3	L3	75	2
Examine appropriate sorting algorithms such as merge sort, heap sort and quick sort etc. based on the problem given with programming examples.	4	L3	75	2
Apply concepts of data structure to develop C++/Python programs for simple applications for solving practical problems in engineering for encouraging employability skills.	5	L4	65	1

D. Program Outcomes and Program Specific Outcomes:

[PO.1.] Engineering knowledge: Apply knowledge of mathematics, natural science, computing, engineering fundamentals and an engineering specialization as specified in WK1 to WK4 respectively to develop to the solution of complex engineering problems.

[PO.2]. Problem analysis: Identify, formulate, review research literature and analyze complex engineering problems reaching substantiated conclusions with consideration for sustainable development. (WK1 to WK4)

[PO.3]. Design/Development of Solutions: Design creative solutions for complex engineering problems and design/develop systems/components/processes to meet identified needs with consideration for the public health and safety, whole-life cost, net zero carbon, culture, society and environment as required. (WK5)

[PO.4]. Conduct investigations of complex problems: Conduct investigations of complex engineering problems using research-based knowledge including design of experiments, modelling, analysis & interpretation of data to provide valid conclusions. (WK8).

[PO.5]. Engineering Tool Usage:: Create, select and apply appropriate techniques, resources and modern engineering & IT tools, including prediction and modelling recognizing their limitations to solve complex engineering problems. (WK2 and WK6).

[PO.6]. The Engineer and The World: Analyze and evaluate societal and environmental aspects while solving complex engineering problems for its impact on sustainability with reference to economy, health, safety, legal framework, culture and environment.(WK1, WK5, and WK7).

[PO.7]. Ethics: Apply ethical principles and commit to professional ethics, human values, diversity and inclusion; adhere to national & international laws. (WK9)

[PO.8]. Individual and Collaborative Team work: Function effectively as an individual, and as a member or leader in diverse/multi-disciplinary teams.

[PO.9].Communication: Communicate effectively and inclusively within the engineering community and society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations considering cultural, language, and learning differences.

[PO.10].Project management and finance: Apply knowledge and understanding of engineering management principles and economic decision-making and apply these to multidisciplinary environments.

[PO.11]. Life-long learning: Recognize the need for, and have the preparation and ability for i) independent and life-long learning ii) adaptability to new and emerging technologies and iii) critical thinking in the broadest context of technological change.(WK8).

[PSO.1]. An ability to understand the concepts of basic Electronics & Communication Engineering and to apply them to various areas like Signal processing, VLSI, Embedded systems, Communication Systems, Digital & Analog Devices, etc.

[PSO.2]. An ability to solve complex Electronics and Communication Engineering problems, using latest hardware and software tools, along with analytical skills to arrive cost effective and appropriate solutions.

[PSO.3]. Wisdom of social and environmental awareness along with ethical responsibility to have a successful career and to sustain passion and zeal for real-world applications using optimal resources as an entrepreneur.

- E. Assessment Plan:** Clearly write the criteria, its description, and associated marks for assessment of student achievements. In addition, the attendance requirements and assignments need to be mentioned in this section.

Criteria	Description	Maximum Marks
Internal Assessment (Summative)	Mid-Term Examination (Close Book)	30
	Class Work Sessional (CWS): Quiz, MOOC, Assignment	30
End Term Exam (Summative)	End Term Exam (Close Book)	40
	Total	100

F. Syllabus:

Introduction to C++/Python: An overview of C++/Python programming language, basic terms and operations. **Linked List:** Representing the linked list in memory with traversing and searching a linked list. **Stack, Queues and Recursion:** Array/ Linked representation of stack and queues with its applications. **Trees:** Tree Definitions, Type of Trees, Traversal Algorithms, **Heaps and Priority Queues:** Heaps, the Natural Mapping, Insertion into a Heap, Removal from a Heap, **Path length:** Huffman's algorithm. Sorting and **Searching:** Various sorting algorithms, Searching and data modification, Hashing. **Graphs:** Types of Graphs, Adjacency and Incidence Matrix for a Graph, Adjacency List for a Graph, Dijkstra's Algorithm, Graph Traversal Algorithms. **Analysis of algorithm:** Synergy between data structures and algorithm, Factors to be considered in the choice of data structures and algorithms.

➤ **TEXTBOOKS**

- [1]. Yashavant Kanetkar, *Data Structures Through C++*, BPB Publications, 2023.
- [2]. Adam Drozdek, *Data Structures and Algorithms in C++*, Cengage Learning, 2012.
- [3]. Michael T. Goodrich, Roberto Tamassia, David M. Mount, *Data Structures and Algorithms in C++*, 2, illustrated Edition, John Wiley C Sons, 2011.
- [4]. J. R. Hubbard, *Data Structures and Algorithms*, Schaum's Outlines. McGraw-Hill, New York, USA, 2000.
- [5]. E. Balagurusamy, *Object Oriented Programming with C++*, McGraw Hill, 2020.
- [6]. Roberto Tamassia, Michael H. Goldwasser, Michael T. Goodrich, *Data Structures and Algorithms in Python*, Wiley; 1st edition, 2013.
- [7]. Mark Allen Weiss, *Data Structures and Algorithm Analysis in C++*, University Paperback, Perason, 2014.
- [8]. Clifford A. Shaffer, *Data Structures C Algorithm Analysis in C++*, Third Edition, Dover Publication US, 2011.

[9]. R. Lewis, L. Deneberg, Data Structures and their Algorithms, Addison-Wesley UK, 1991.

G. Lecture Plan: The lecture plan of the course indicates clear schedule of content delivery, session outcome, corresponding CO, mode of delivery, and mode of assessment. The general template of lecture plan is mentioned as follows:

Lecture No.	Topic	Session Outcome	Corresponding CO	Mode of delivery	Mode of assessing CO
1.	Introduction and Course Hand-out briefing	Understand basics of subject and assessment	1	Lecture-based teaching-learning	NA
2.	Introduction to C++/Python, Features, Program Structure	Understand basic language syntax and operations	1	Lecture based teaching-learning	InClass Quiz/ Mid Term/End Term
3.	Basic Terms: Variables, Data Types, Operators, Input/Output	Understand basic language syntax and operations	1	Lecture based teaching-learning	InClass Quiz/ Mid Term/End Term
4.	Control Statements and Functions (Overview)	Understand basic language syntax and operations	1	Lecture based teaching-learning	InClass Quiz/ Mid Term/End Term
5.	Introduction to Data Structures and Memory Representation	Define basic data structures and explain their memory allocation	1	Lecture based teaching-learning	InClass Quiz/ Mid Term/End Term
6.	Linked List: Concept, Types, Node Representation	Represent linked lists in memory	1	Lecture based teaching-learning	InClass Quiz/ Mid Term/End Term
7.	Linked List Operations: Creation and Insertion, Deletion and Traversal	Represent linked lists in memory	1	Lecture based teaching-learning	InClass Quiz/ Mid Term/End Term
8.	Searching in Linked List, Applications of Linked List, Numerical Problems and Practice on Linked List	Perform traversal and search operations	1	Lecture based teaching-learning	InClass Quiz/ Mid Term/End Term
9.	Tutorial	Perform traversal and search operations	1	Problem solving/Tutorial	InClass Quiz/ Mid Term/End Term
10.	Class Test	Assess understanding and application of covered topics	1	Problem solving/Tutorial	Quiz
11.	Stack: Concept and array representation	Implement stacks and queues in memory	2	Lecture based teaching-learning/	Quiz/ Mid Term/End Term
12.	Stack using Linked List, Stack Application	Implement stacks and queues in memory	2	Lecture based teaching-learning	Quiz/ Mid Term/End Term
13.	Queue: Concepts and Array Representation	Implement stacks and queues in memory	2	Lecture based teaching-learning	Quiz/ Mid Term/End Term

14.	Queue using Linked List and Circular Queue	Implement stacks and queues in memory	2	Lecture based teaching-learning	Quiz/ Mid Term/ End Term
15.	Queue Applications	Implement stacks and queues in memory	2	Lecture based teaching-learning/	Quiz/ Mid Term/ End Term
16.	Recursion and Recursive Algorithms	Apply recursion and linear DS to solve problems	2	Lecture based teaching-learning	Quiz/ Mid Term/ End Term
MID SEMESTER EXAMINATION					
17.	Trees: Terminology and Types	Differentiate between various tree structures	3	Lecture based teaching-learning	InClass Quiz/ End Term
18.	Binary Trees and Binary Search Trees	Differentiate between various tree structures	3	Lecture based teaching-learning	InClass Quiz/ End Term
19.	Tree Traversal Algorithms	Implement in-order, pre-order, and post-order traversals	3	Lecture based teaching-learning	InClass Quiz/ End Term
20.	Red-Black Trees	Implement in-order, pre-order, and post-order traversals	3	Lecture based teaching-learning	InClass Quiz/ End Term
21.	Red-Black Trees+ Student Presentation	Assess understanding and application of covered topics	3	Individual learning	InClass Quiz/ End Term
22.	Heap: Concepts and Natural Mapping	Understand heap properties and priority queue concepts	3	Lecture based teaching-learning	InClass Quiz/ End Ter
23.	Heap Operations and Priority Queue	Understand heap properties and priority queue concepts	3	Lecture based teaching-learning	InClass Quiz/ End Term
24.	Huffman's Algorithm	Execute heap operations and calculate path lengths	3	Lecture based teaching-learning	InClass Quiz/ End Term
25.	Huffman's Algorithm +Class Test	Assess understanding and application of covered topics	2,3	Learning through problem-solving	Quiz
26.	*Bubble, Selection and Insertion Sort	Select and apply appropriate sorting/searching methods	4	Lecture based teaching-learning	InClass Quiz/ End Term
27.	*Merge Sort	Select and apply appropriate sorting/searching methods	4	Lecture based teaching-learning	InClass Quiz/ End Term
28.	*Heap Sort	Select and apply appropriate sorting/searching methods	4	Lecture based teaching-learning	InClass Quiz/ End Term
29.	*Quick Sort	Select and apply appropriate sorting/searching methods	4	Lecture based teaching-learning	InClass Quiz/ End Term

30.	*Searching Algorithms and Hashing	Select and apply appropriate sorting/searching methods	4	Lecture based teaching-learning	InClass Quiz/ End Term
31.	Various sorting algorithms,	Implement hash tables and resolve collisions	4	Lecture based teaching-learning	InClass Quiz/ End Term
32.	Presentation on Tree by student groups	Construct matrix representations of graphs	4	Group- teaching and learning	InClass Quiz/ End Term
33.	Presentation on searching Algorithms by student groups	Identify the appropriate basic data structure for a given problem and explain its corresponding memory footprint	4	Group- teaching and learning	InClass Quiz/ End Term
34.	Presentation on sorting Algorithms by student groups	Identify the appropriate basic data structure for a given problem and explain its corresponding memory footprint	4	Group- teaching and learning	InClass Quiz/ End Term
35.	Graph Representation	Introduction of basics of graph	5	Lecture based teaching-learning	InClass Quiz/ End Term
36.	Graph Traversal: BFS and DFS	Implement BFS and DFS traversals	5	Lecture based teaching-learning	InClass Quiz/ End Term
37.	Dijkstra's Algorithm and Algorithm Analysis	Calculate the shortest path in a weighted graph	5	Lecture based teaching-learning	InClass Quiz/ End Term
38.	Graph Traversal Algorithms	Calculate the shortest path in a weighted graph	5	Lecture based teaching-learning	InClass Quiz/ End Term
39.	Synergy between data structures and algorithm	Analyze efficiency and select appropriate data structures	5	Lecture based teaching-learning	InClass Quiz/ End Term
40.	Factors to be considered in the choice of data structures and algorithms	Analyze efficiency and select appropriate data structures	5	Lecture based teaching-learning	InClass Quiz/ End Term
41.	Presentation on graphs traversal by student groups	Analyze efficiency and select appropriate data structures	5	Group- teaching and learning	InClass Quiz/ End Term
42.	Class test	Identify the appropriate basic data structure for a given problem and explain its corresponding memory footprint	4,5	Lecture based teaching-learning	InClass Quiz/ End Term

H. Course Articulation Matrix:

CO	STATEMENT														
		PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PSO 1	PSO 2	PSO 3
1	Explain concepts of object-oriented programming using C++/Python.	3	3	3	3	2							2		
2	Explain the basic operations on arrays, lists, stacks and queue data structures with their programming examples.	3	3	3	3	2							1		
3	Develop the notions of trees, binary search trees, Red Black tree, Heap with their programming examples.	3	3	3	2	2							1		
4	Examine appropriate sorting algorithms such as merge sort, heap sort and quick sort etc. based on the problem given with their programming examples.	3	3	3	2	2						1	1		
5	Apply concepts of data structure to develop C++ programs for simple applications for solving practical problems in engineering for	3	3	3	2	2						2	1	1	

1: Low Correlation; 2: Moderate Correlation; 3: Substantial Correlation